

Jharkhand University of Technology, Ranchi

Diploma 6th Semester Examination, 2025

Subject : Cloud Computing (Elective-III)

Subject Code : CSE 607

Time Allowed : 3 Hours

Full Marks : 80

Pass Marks : 26

*Answer in your own words. Answer five questions in which
Question No. 1 is compulsory. The figures in the margin indicate full marks.
All questions carry equal marks.*

1. Choose the correct option for the following questions (Answer any eight):

2×8=16

(i) Which of the following best describes cloud computing?

- (a) Computing using the internet
- (b) Distributed computing over a network
- ✓(c) On-demand network access to share resources
- (d) Computing without servers

(ii) Which service model provides virtualized computing resources over the internet?

- (a) SaaS
- (b) PaaS
- ✓(c) IaaS
- (d) DBaaS

(iii) Cloud storage provides

- (a) direct hardware access.
- (b) temporary cache memory.
- (c) scalable and remote data storage.
- ✓(d) optical storage solutions.

(iv) Which component acts as the user interface in cloud computing?

- (a) Server hardware
- (b) Data centre
- ✓(c) Client device
- (d) Hypervisor

(v) Which model provides the user with control over operating systems and storage?

- (a) SaaS
- (b) FaaS
- ✓(c) IaaS
- (d) PaaS

(vi) What is a major advantage of SaaS?

- (a) Full control over data centres
- ✓(b) Easy software upgrades and maintenance
- (c) Hardware customization
- (d) Manual installation

(vii) Which of the following is a trend in cloud service provisioning?

- (a) Manual resource allocation (b) Decentralized storage only
(c) Auto-scaling and serverless architectures- (d) On-premise only computing

(viii) Which of the following enables automatic scaling in cloud services?

- (a) Cloud scheduler (b) Load testing
(c) Elastic load balancer (d) Static provisioning

(ix) Google Calendar is an example of

- (a) PaaS (b) FaaS
(c) IaaS (d) SaaS

(x) Which of these is an open-source cloud platform?

- (a) Azure (b) Google Cloud
(c) OpenNebula (d) Salesforce

2. (a) Explain the evolution of cloud computing from distributed computing. Highlight key differences.

(b) Differentiate between the three service models of cloud computing: IaaS, PaaS and SaaS with suitable examples. 8+8

3. (a) Discuss the importance of cloud security from a service perspective. What are the key concerns and solutions?

(b) Explain how client system interact with cloud platform in a service-based environment. 8+8

4. (a) Illustrate the service deployment lifecycle in IaaS, PaaS and SaaS models.

(b) Discuss the recent trends in service provisioning with cloud computing. 8+8

5. (a) What are the benefits and challenges of using cloud-based collaborative tools in organizations?

(b) Explain the role of cloud services in project management applications. 8+8

6. (a) Compare the features of OpenNebula and Eucalyptus as open-source cloud platforms.

(b) Analyze how open-source clouds are influencing the cloud computing market. 8+8

7. Write short notes on *any four* of the following:

4x4

(a) Cloud Storage

(b) Cloud-based calendar service

(c) Microsoft Azure

(d) Limitations of open-source cloud platforms

(e) Cloud computing architecture